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Experiment 1: Case Study

Problem Statement: Choose a real-world case study on KL ERP System, create user stories, select an appropriate software development process model, identify a suitable programming language, and propose an implementation and evaluation strategy.

Real-World case study a KL ERP System

1. Functional Requirements

Customer

- Register and login
- Search products
- View product details
- Add products to cart
- Place orders
- Make online payments
- Track order status
- View purchase history
- Submit product reviews

Admin

- Manage products
- Manage categories
- Update inventory
- View customer details
- Process orders
- Generate sales reports

2. User Stories

USER ID	User Story	Priority
FA-01	As a faculty member, I want to view and manage my teaching timetable so that I can plan classes effectively.	High 8
FA-02	As a faculty member, I want to upload assignments and grade them online so that students receive timely feedback.	High 6
FA-03	As a faculty member, I want to send announcements or messages to my class so that students stay informed.	High

USER ID	User Story	Priority
FA-04	As a faculty member, I want to record attendance so that student attendance can be tracked accurately.	High
FA-05	As a faculty member, I want to enter internal and examination marks so that the student can view their academic performance	Medium3
FA-06	As a faculty member, I want to upload course materials so that students can access learning resources.	High8
FA-07	As a faculty member, I want to post assignments so the students know their academic tasks.	High6

Administrator User Stories

User ID	User Story	Priority
AUS-01	As an administrator, I want to generate and manage institutional timetables so that classes are scheduled without conflicts.	High
AUS-02	As an administrator, I want to oversee fee collection and financial records so that institutional accounts remain accurate.	High
AUS-03	As an administrator, I want to schedule examinations and publish results so that academic processes run smoothly.	High
AUS-04	As an administrator, I want to manage user roles and permissions so that system security is maintained.	High
AUS-05	As an administrator, I want to manage student records so that all student information remains updated.	High
AUS-06	As an administrator, I want to generate reports so that institutional performance can be monitored	High
AUS-07	As an administrator, I want to assign faculty members to course so that teaching responsibilities are organized.	Medium

3. Selected Software Development Process Model

Agile Scrum

Reason for Selection

- Requirements change frequently across departments.
- Continuous feedback from students, faculty, and administrators.
- Incremental delivery of ERP modules.
- Faster development cycles with sprint reviews.
- Easy bug fixing and adaptability.
- Well-suited for web-based academic applications.

Sprint Plan

Sprint	Features
Sprint 1	User Registration & Login
Sprint 2	Course Registration Module
Sprint 3	Attendance Tracking (Faculty entry + Student view)
Sprint 4	Examination Management (marks entry, results publishing)
Sprint 5	Payment Gateway
Sprint 6	Timetable Scheduling & Faculty Assignment
Sprint 7	Reports & Admin Dashboard
Sprint 8	Testing (Unit, Integration, UAT) & Deployment
Sprint 9	Maintenance & Continuous Improvement

4. Programming Language Selection

Backend

- Java (Spring Boot)

Reasons

- Secure
- Scalable
- Enterprise-level applications
- REST API support
- Excellent database connectivity

Frontend

- HTML5
- CSS3

- JavaScript
- React.js

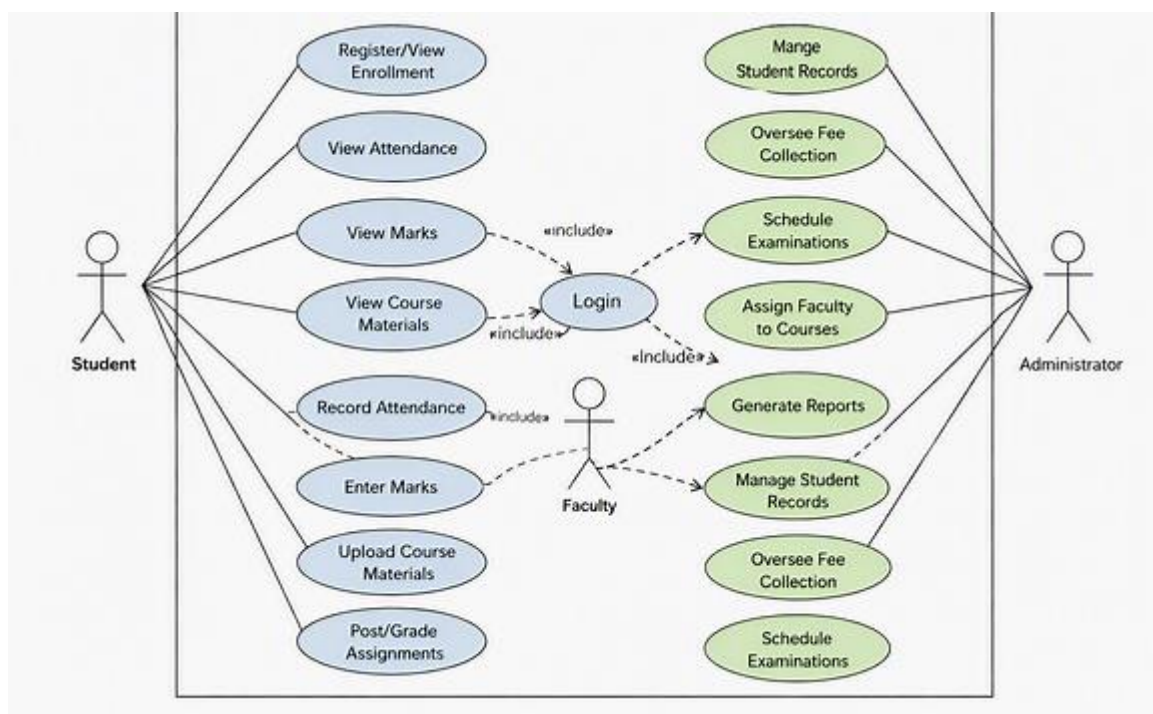
Database

- MySQL

Other Technologies

- REST API
- JWT Authentication
- Stripe/Razorpay Payment Gateway
- Git
- Docker
- AWS Cloud

5. System Architecture



Phase 1 – Requirement Analysis

- Gather requirements from students, faculty, and administrators.
- Create the Software Requirement Specification (SRS) document.
- Prepare use case diagrams for each module (Student, Faculty, Admin).

Phase 2 – System Design

- Design database schema for student, course, attendance, exam, and fee modules.
- Create UI wireframes for dashboards and forms.
- Develop UML diagrams (class, sequence, activity).
- Define system architecture using Spring Boot + React + MySQL.

Phase 3 – Development

- Implement user authentication and role-based access control.
- Develop course registration and timetable modules.
- Build attendance and examination management features.
- Integrate online fee payment and report generation.
- Create admin dashboard for institutional monitoring.

Phase 4 – Testing

Unit Testing

- Login
- Registration
- Course registration
- Attendance tracking

Integration Testing

- Attendance + Marks entry
- Fee payment + Reports

System Testing

- End-to-end ERP workflow validation

User Acceptance Testing (UAT)

- Verify academic and administrative requirements with real users.

Phase 5 – Deployment

- Deploy using Docker containers.
- Host on AWS EC2 for scalability.
- Use MySQL database for persistent storage.
- Configure Nginx web server for load balancing and security.

7. Evaluation Strategy

Evaluation Parameter Method

Functional Accuracy	Requirement verification and module validation
Performance	Load testing using JMeter
Security	Authentication, authorization, and penetration testing
Reliability	Continuous monitoring and backup validation
Usability	User feedback from students, faculty, and admin
Scalability	Concurrent user testing under peak load
Maintainability	Code review, documentation, and version control (GitHub)

10. Expected Outcomes

- Centralized academic and administrative management.
- Secure and scalable ERP platform.
- Faster course registration and result processing.
- Reduced manual workload and paperwork.
- Improved communication among students, faculty, and administrators.
- Reliable data storage and reporting.
- Continuous enhancement through Agile iterations.